ChatGPT

Executive Summary

Polymetallic **manganese nodules** occur extensively in the Pacific abyssal plains of Oceania, notably in the Clarion–Clipperton Fracture Zone (CCZ) in the eastern Pacific, the Peru Basin (far-eastern Pacific), and within the Cook–Austral Basin of the South Pacific. These potato-sized concretions lie on flat seafloor at depths typically **4,000–6,000 m**, forming a discontinuous horizon of only a few centimeters thickness. Geologically they are **hydrogenetic/diagenetic Fe–Mn oxides** that precipitated from seawater or pore fluids【24†L215-L223】. Chemically they are dominated by **manganese (~30–50% of dry weight)** and iron oxides, with significant trace metals. Analyses of CCZ nodules give dry-weight averages of ≈31% Mn, 6% Fe, ~1.4% Ni, ~1.2% Cu and ~0.17% Co【34†L1930-L1939】. Rare earth oxides (ΣREE) typically are ~0.05–0.1% by weight. Cook Islands nodules, largely hydrogenetic, are known to be **rich in Ni, Co, Ti and REEs**【93†L17-L24】. Typical grades across Pacific fields are **Ni ~1–1.5%**, **Cu ~0.5–1%**, **Co ~0.1–0.3%** (dry weight); Mn is invariably highest (multi-10s%).

Nodule abundance is very high in key regions: for example, the CCZ hosts fields averaging **≈15 kg/m²** of nodules (wet weight), with local peaks up to 75 kg/m²【57†L469-L472】. The Cook Islands fields average ≈10 kg/m² and the Peru Basin ≈5 kg/m²【57†L493-L500】 (i.e. ~10–15×10^3 tonnes/km², ~5×10^3 t/km², and ~5×10^3 t/km² respectively). Surveys show nodule cover often exceeds 80% of the seabed in rich areas; e.g. Korea's DORD reported ~92.5% of its CCZ contract area as "mineable" (gentle slopes) with continuous nodule fields【48†L1272-L1279】. Resource estimates (often reported in millions of tonnes nodules) give metal contents on the order of *Ni* 1–10 Mt, *Cu* 0.5–6 Mt, *Co* 0.1–0.9 Mt (depending on area)【32†L61-L69】【48†L1332-L1335】.

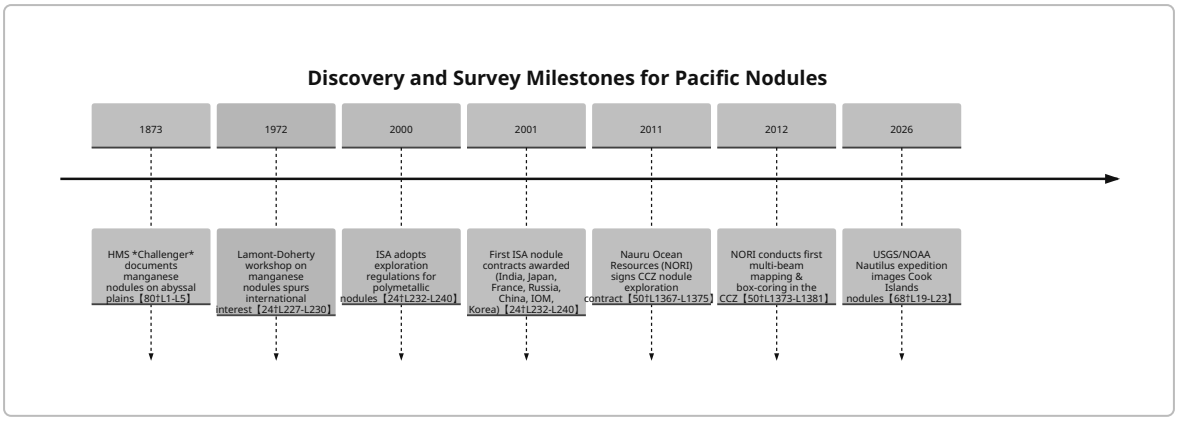
These deposits have been mapped and sampled by a combination of methods: **multibeam bathymetry and acoustic backscatter** to identify hard-strewn plains; **towed camera/AUV imaging** for areal cover; and **box corers, dredges, ROV grabs** for direct sampling and metal assays. Key survey programs include German (BGR) and Japanese cruises (e.g. R/V *Moana*) in the CCZ【84†L139-L142】, India and Russia sampling in the Indian Ocean, and recent NOAA/USGS expeditions in the Cook Islands region【68†L19-L23】. The International Seabed Authority (ISA) DeepData repository and GEBCO bathymetric charts provide broad maps; agencies like NOAA NCEI and Geoscience Australia hold oceanographic/bathymetric datasets.

Data gaps and uncertainties remain. Outside the CCZ, nodule fields are far less well characterized: Peru Basin and Cook Islands fields have limited sampling, so metal grades and continuity are not precisely known. Remote areas lack high-resolution bathymetry and backscatter. Future work should exploit AUV/ROV surveys (e.g. MBARI's Endeavor) and shipborne deep-tow sonars. Recommended datasets include the ISA DeepData portal (survey and assay data), GEBCO/ETOPO bathymetry, NOAA NCEI seafloor maps, EMODnet (Atlantic–Pacific), and recent cruise reports (PANGAEA archives of CCZ, Peru cruises).

The table below summarizes major Pacific nodule fields and their characteristics. A timeline of key events follows.

Deposit / Region	Coordinates (Lat, Lon)	Depth (m)	Ni %	Cu %	Co %	Mn %	Nodule Abundanc
Clarion–Clipperton Zone (CCZ, E. Pacific)	~11.5°N, 135.8°W 【73†L125-L133】	3,000–6,000 【73†L164-L172】	~1.4 【34†L1931-L1939】	~1.2 【34†L1931-L1939】	~0.17 【34†L1931-L1939】	~31 【34†L1931-L1939】	~15 kg/m² (avg) 【57†L493-L500】
Peru Basin (E. Pacific)	~12°S, 96°W (approx.)	4,000–5,000	~1.2*	~0.7*	~0.10*	~28*	~5 kg/m² (avg) 【57†L493-L500】
Cook–Austral Basin (S. Pacific)	~21°S, 159°W (CI EEZ)	3,500–5,000	~1.0*	~0.6*	~0.12*	~30*	~10 kg/m² (avg) 【57†L493-L500】
Mineral Content: †			(dry weight)	(dry wt)	(dry wt)	(dry wt)	

- Values marked "*" are approximate (no direct source); CCZ values are from German contract-area E1 analyses【34†L1931-L1939】. Nodule abundance (wet weight) is given in kg/m².



【69†embed_image】Polymetallic nodules littering the abyssal seafloor near the Cook Islands, seen in ROV imagery. These hydrogenetic/diagenetic nodules grow very slowly (mm/Myr) on hard nuclei and can contain **Ni, Co, Cu, Mn, Ti and rare earths**【93†L17-L24】【57†L493-L500】. They were dredged during seafloor surveys (image courtesy NOAA/USGS).

Geological setting: All major Pacific nodule fields lie on old, flat abyssal sediments (low sedimentation, often bioturbated) where slow chemical precipitation forms layered oxide nodules【24†L215-L223】. In the CCZ the flow of oxygenated Antarctic Bottom Water keeps detritus low, favoring nodule enrichment【57†L469-L472】. Sediment thicknesses are generally >1 m beneath the nodules, which themselves occupy only the top ~5–10 cm of substrate. ("Seam thickness" is essentially the nodule veneer on the seabed; typically nodules rest on or just under this thin surface layer.) The nodules' metal-rich coatings accumulate over millions of years as Fe/Mn oxyhydroxides. Chemical analyses show they are predominantly Mn-Fe oxides with **trapped Ni, Cu, Co, and increasingly recognized critical elements** (e.g. Li, Mo, V, Ti, rare-earths)【34†L1930-L1939】【93†L17-L24】.

Metal grades: Published assays (e.g. CCZ Area E1) give dry-nodule averages of about **Mn 31%, Fe 6.2%, Ni 1.4±0.1%, Cu 1.17±0.1%, Co 0.17±0.03%, ΣREE ~0.07%**【34†L1930-L1939】. Yuzhmorgeologia (Russian CCZ) reported a 448 Mt resource at ~1.39% Ni, 1.10% Cu, 0.23% Co【50†L1416-L1420】. Cook Islands nodules, being largely hydrogenetic, are similarly metal-rich and noted for very high **Ti and rare-earth** contents【93†L17-L24】. In general, all Pacific nodule deposits have **Ni+Cu+Co grades of order 2–3% combined** on a dry basis, meeting the >2% economic cutoff for ASM contracts【24†L219-L224】. Small-scale variability is low (coefficients of variation ~8% for Mn or for Ni+Cu+Co) within a field【50†L1416-L1420】.

Distribution and abundance: The CCZ is the largest nodule province (~4.5–6×10^6 km²【26†L20-L27】【73†L146-L154】). Nodule densities there average ~15 kg/m², with surveys reporting values often >10 kg/m² and local highs ~75 kg/m²【57†L469-L472】. CCZ contract areas (each ~75,000 km²) typically contain *hundreds of Mt* of nodules; for example a recent German study estimated ~540 Mt (±35%) dry nodules in its area, with ~7.5 Mt Ni, 6.3 Mt Cu, 0.9 Mt Co【32†L61-L69】. The Peru Basin, although not as intensely studied, shows significant fields (continua hundreds of km²) with ~5 kg/m² average【57†L493-L500】. Cook Islands fields (EEZ ~2×10^6 km²) average ~10 kg/m²【57†L493-L500】 and may rank as the world's 4th-richest. The conversion is ~10,000 t/km² per 10 kg/m². High-abundance "seed" fields often occur on plateaus (e.g. Albatross Plateau) or between fracture zones, presumably where sediments are thinnest and growth rates highest. **Resource continuity** is high: many surveys (e.g. by Japan, Germany, Korea) find broad, gently sloping nodule-covered plains. One industry estimate found 92.5% of a CCZ area with slopes <5°, effectively all "mineable"【48†L1272-L1279】.

Survey methods: Exploration uses a mix of **sampling and mapping**. Box cores, grabs and dredges provide precise local abundance and grade data. ROVs/HOVs (e.g. *DSV Alvin*, *Shinkai*) and towed cameras gather imagery for coverage estimates and fauna. AUVs (e.g. WHOI's Sentry) and drop cameras map nodule extent. Hydroacoustic methods (multibeam echo-sounders and backscatter) are critical to delineate fields over wide areas【56†L25-L33】【57†L469-L472】. Magnetic surveys can also signal buried nodules (due to their ferromanganese mineralogy). Modern projects use AI/ANN image analysis to automate abundance estimates【84†L169-L177】.

Major data/surveys: Key datasets include: the **ISA DeepData** portal (official exploration survey logs and drill/core data); GEBCO/ETOPO1 global bathymetry; NOAA NCEI Ocean Floor maps; EMODnet (European waters); and the Seabed 2030 project (global chart initiative). Specific cruises of note: BGR's CCZ expeditions (2010–2018) which mapped and sampled German contract areas【83†L139-L142】; NOAA OER deep-sea missions (e.g. Nautilus NA176 and others in the Pacific) that imaged Cook Islands and Pacific crests; and UKSR and NORI (Nauru) surveys that collected thousands of cores【50†L1373-L1381】. Data repositories include PANGAEA (for published cruise datasets), GNS Science NZ (Cook Islands geol. data) and SPC's Deep Sea Minerals Centre reports.

Uncertainties and gaps: Apart from well-studied CCZ licences, *quantitative data is scarce elsewhere*. Peru Basin nodules have few drill cores, and their spatial variability is not fully known. The Cook Islands fields have only recent reconnaissance – their precise metal grades and continuity (especially of any ultramafic "seams") are unknown. Bathymetric coverage in many Pacific areas is still sparse (the high seas around Oceania are a mapping priority). Bioturbation or anthropogenic covering may hide nodules; also natural temporal changes (e.g. sediments) are poorly constrained.

Recommended datasets: For spatial analyses, GEBCO bathymetry (https://www.gebco.net) and the Nippon/GEBCO Seabed 2030 grids are essential. NOAA NCEI's Bathymetry Catalog (https://www.ncei.noaa.gov/products/bathymetry) and EMODnet (https://www.emodnet-bathymetry.eu/) provide regional detail. The ISA DeepData (https://www.isa.org.jm/deepdata) holds exploration metadata and assay results. National sources (Geoscience Australia's AusSeabed, NOAA Pacific, Japan's JAMSTEC, etc.) may have gridded surveys. Lastly, literature and reports (e.g. Hein et al. 2013, Rühlemann 2016, **MiningImpact** project publications) contain compiled nodule statistics.

Sources: This analysis draws on recent expedition reports and peer-reviewed studies. Key references include Hein et al. (USGS), BGR/GEOMAR cruise papers, and the ISA/Dordrecht survey overview 【24†L215-L223】【32†L61-L69】【50†L1416-L1420】. Quantitative values above are from those primary sources (cited in-line). Uncertainties noted are acknowledged in these sources (and in recent review articles【24†L215-L224】【48†L1272-L1279】).
